

DIW Data Narrator Documentation

Repository

GitHub Repo: <https://github.com/harshith-byte/HMI-Project>

Overview

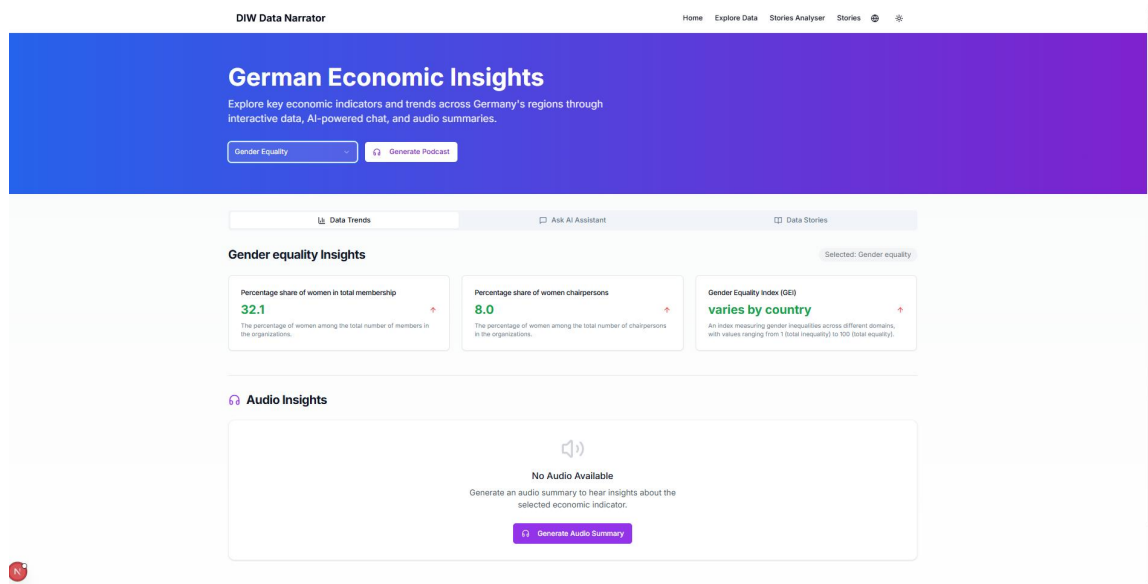
DIW Data Narrator is a web application designed to explore, analyze, and narrate German economic data using interactive visualizations, chatbot summaries, and AI-generated stories. It empowers users to gain insights through economic indicators across multiple domains using a user-friendly interface and story-driven data exploration.

Project Walkthrough with Screenshots

This section presents a visual walkthrough of the DIW Data Narrator application through sequential screenshots.

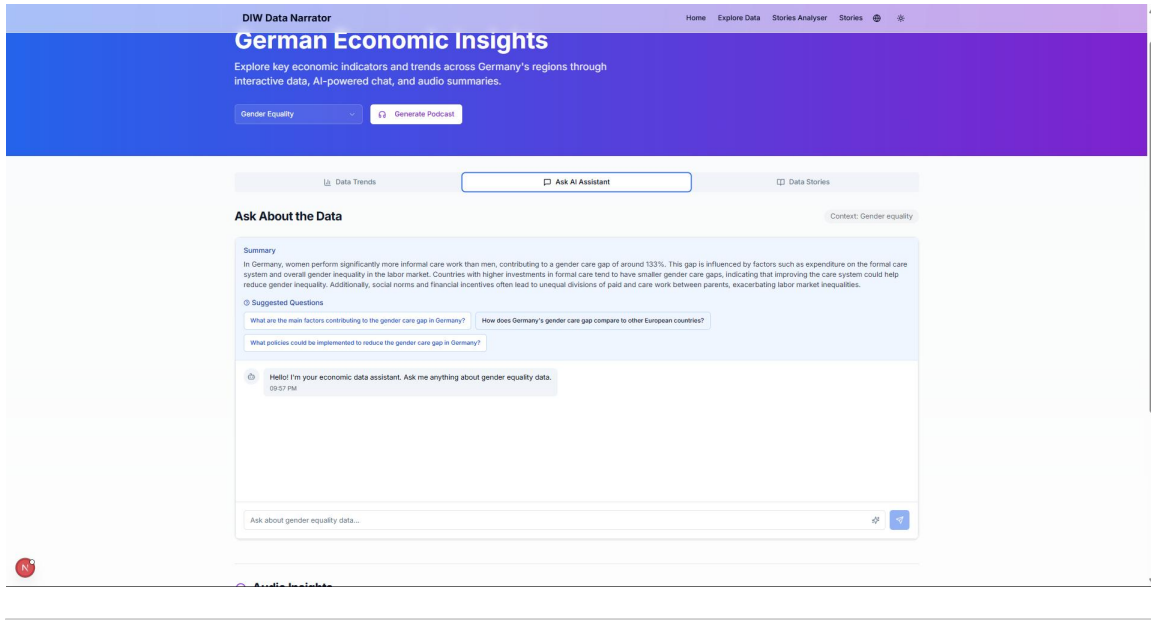
1. Home Page – Domain and Indicator Selection

- Users can select a domain from a dropdown (e.g., Labour, Gender Equality).
- Once selected, important indicators for the domain are displayed, such as job vacancy rate and participation rate. Key trends of the domain
- There is a button to generate a podcast summary of the insights.



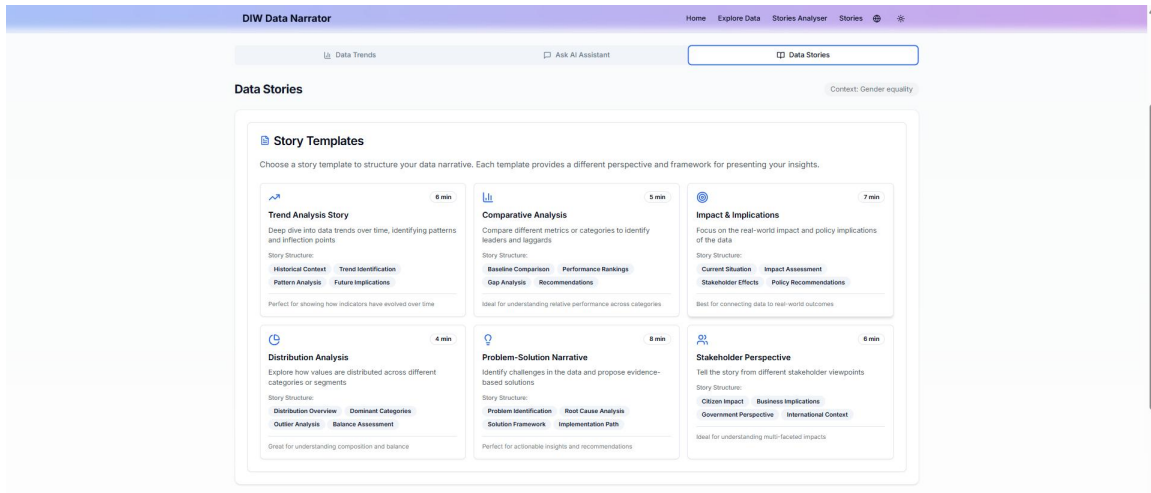
2. Ask AI Assistant – Contextual Summary & Chat

- After domain selection, users can switch to the “Ask AI Assistant” tab.
- It displays a contextual summary about the selected domain.
- Suggested follow-up questions are also shown.
- A chatbot interface allows users to interact with the assistant.



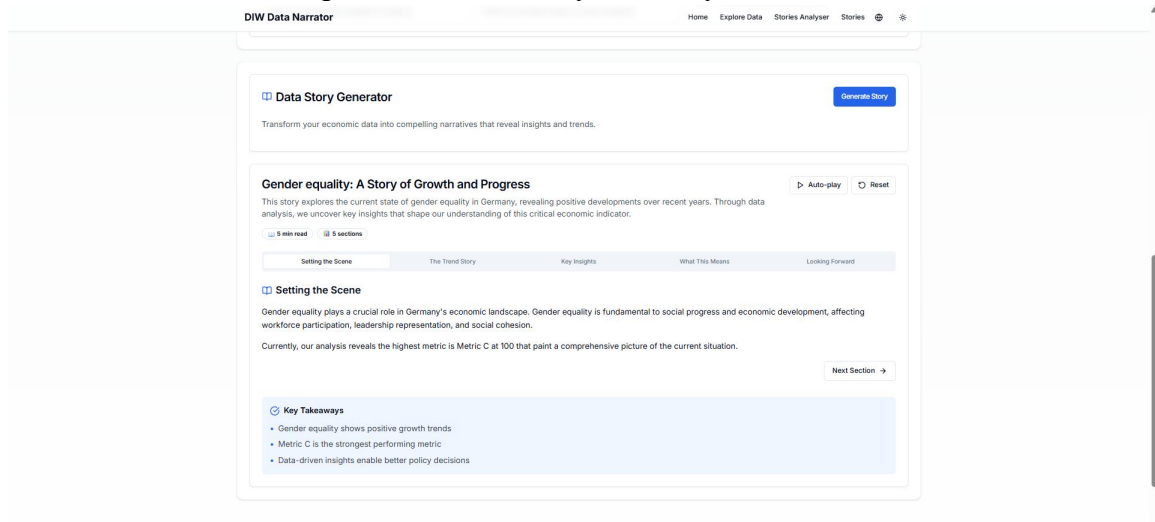
3. Story Generator – Choose a Template

- From the “Data Stories” section, users select a template to generate a story.
- Templates include: Trend Analysis, Comparative Analysis, Impact & Implications, etc.
- Each template outlines the structure and estimated reading time.
- Example of a story generated using the “Trend Analysis” template.
- Story titled “Labour: A Story of Growth and Progress”.
- Structured into multiple sections with key takeaways.



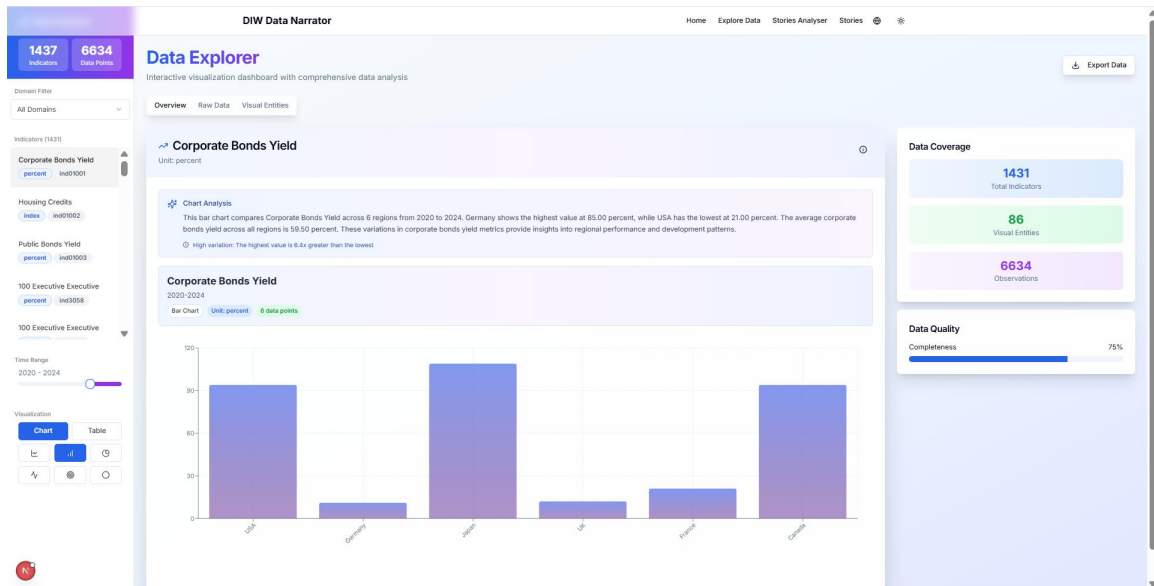
4. Generated Options Story

- Example of a story generated using the “Trend Analysis” template.
- Story titled “Labour: A Story of Growth and Progress”.
- Structured into multiple sections with key takeaways.



5. Explore Data – Indicator Dashboard

- An interactive dashboard to browse and visualize all available indicators.
- Users can apply domain filters, select time ranges, and choose visualization types (Chart/Table).
- A bar chart for “Corporate Bonds Yield” is shown as an example.



6. Research Stories – Cross-Domain Analysis In Story Analyser

- A collection of published stories from various domains.
- Filter and search bar to browse stories by keyword, author, or domain.
- Story cards display summaries and tags.

The screenshot displays the 'Research Stories' section of the 'DIW Data Narrator'. The interface includes a search bar and filters for Category, Author, Sort By, and View. Below the filters, there are three story cards: 'The Electric Freight Gambit — Germany's Trucking Sector at a Crossroads', 'The Toddler Gap: Why Family Policy Is Still Failing Parents', and 'Unequal Burdens: Gender Gaps in a Reshaped Economy'. Each card provides a brief summary and tags for exploration.

Filter & Search Stories

Search stories by title, subtitle, or content...

Category: All Categories | Author: All Authors | Sort By: Newest First | View: Grid | List

Showing 17 of 17 stories

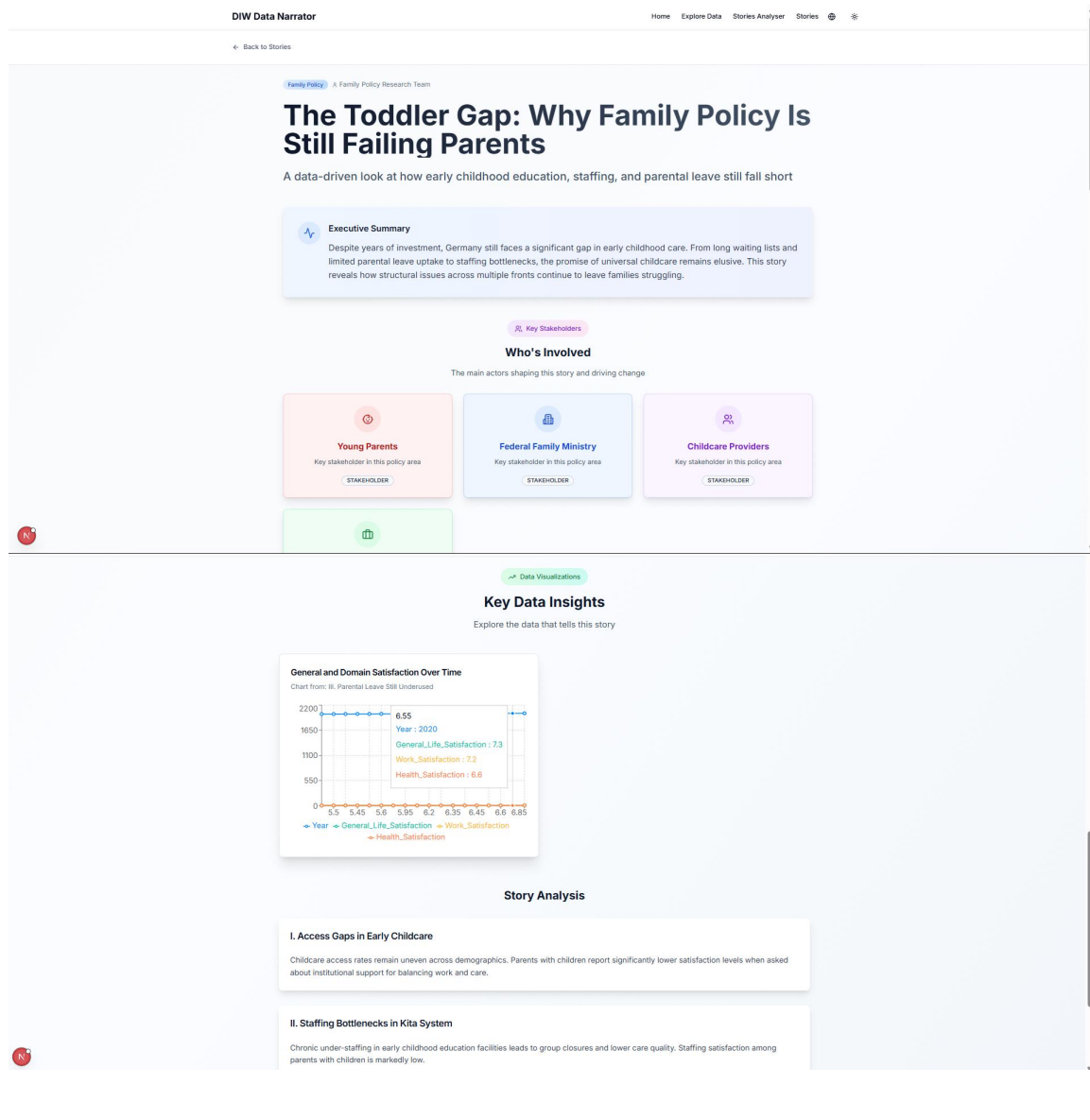
The Electric Freight Gambit — Germany's Trucking Sector at a Crossroads
How Germany's Freight Transport Sector Is Struggling to Decarbonize Under Pressure
Freight transport accounts for 28% of Germany's transport emissions. As diesel costs surge under rising carbon prices and battery-electric truck sal...
Logistics companies | Truck manufacturers
+1
Explore →

The Toddler Gap: Why Family Policy Is Still Failing Parents
A data-driven look at how early childhood education, staffing, and parental leave still fall short
Despite years of investment, Germany still faces a significant gap in early childhood care. From long waiting lists and limited parental leave uptake to...
Young Parents | Federal Family Ministry | +2
Family Policy Research Team | Explore →

Unequal Burdens: Gender Gaps in a Reshaped Economy
Tracing how structural shifts and crises have intensified the economic divide for women
The economic aftermath of the COVID-19 pandemic and the ongoing cost-of-living pressures have exacerbated existing gender disparities in income...
Women in executive roles | Corporate boards
+1
Socioeconomic Analysis Team | Explore →

7. Story Detail – Example: Toddler Gap

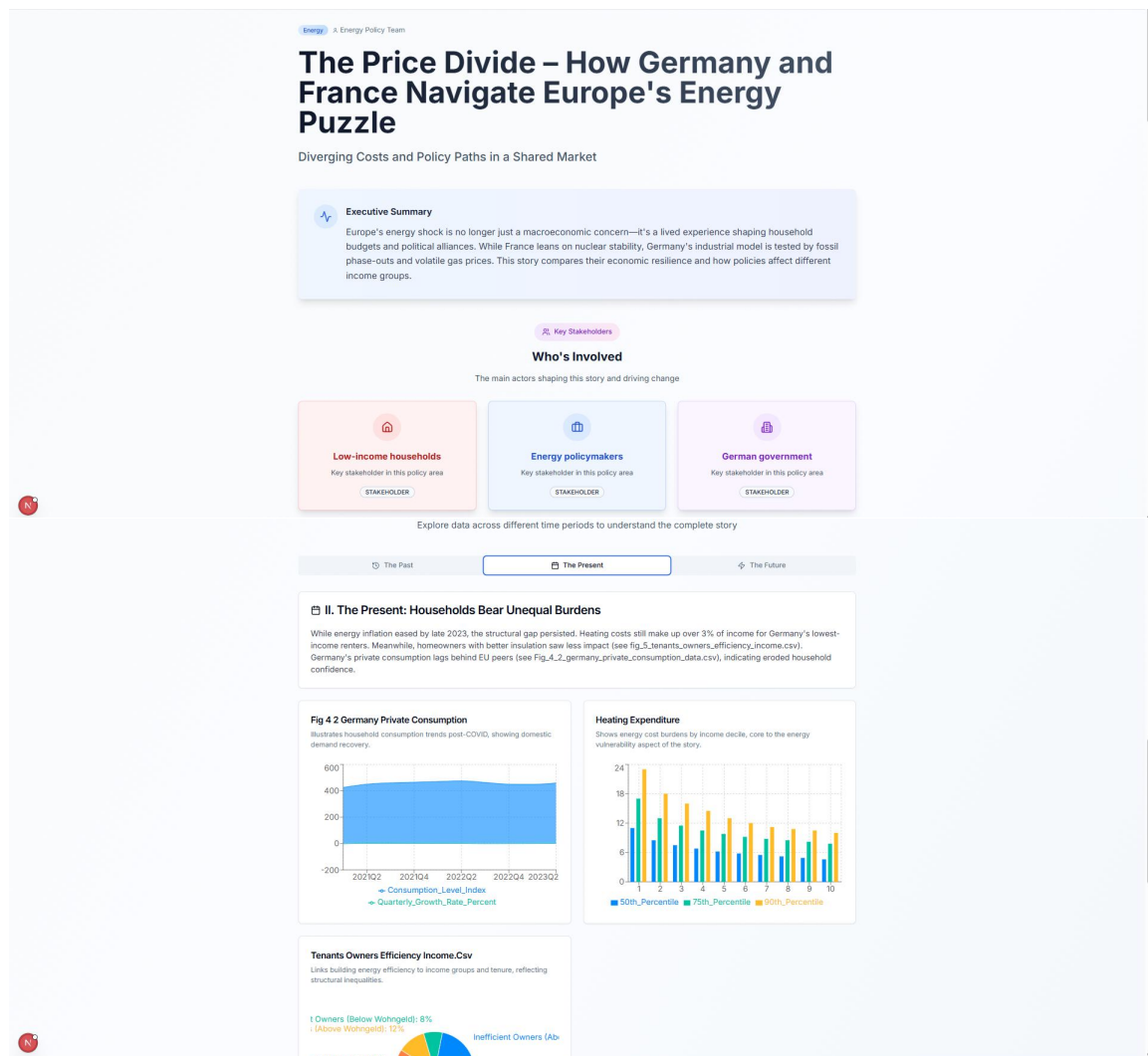
- Detailed view of a cross-domain story.
- Includes an executive summary and stakeholder view.
- Key actors like Young Parents and Family Ministry are highlighted.



8. Timeline View – Story in 3 Parts for example The Price Divide – How Germany and France Navigate Europe's Energy Puzzle

- Also some story have Timeline breakdown of a story: Past, Present, and Future.
- Visuals support each period.

- Example: Heating burden and private consumption in present timeline.



10. Domain-Specific Story List – for example Macroeconomy

- Shows all macroeconomic stories and visualisations
- Each includes read time, complexity, tags, and publication date.

DIW Data Narrator

HomeExplore DataStories AnalyserStories

Story Domains

Explore stories across different domains

124161

Total StoriesDomainsMinutes

Search domains...Filter

Featured Domain

Macroeconomics

Economic trends, policy analysis, and market dynamics

45min4

4 stories

Explore

All Domains

Macroeconomics

Economic trends, policy analysis, and market dynamics

45min4

4 stories

Explore

Energy

Mobility trends, infrastructure, and sustainable transport

60min4

4 stories

Explore

Gender Equality

Gender gaps, workplace equality, and social progress

42min3

3 stories

Explore

Labour

Structural shifts in European employment post-COVID, including hybrid work, sectoral divergence, and reskilling challenges

1min1

1 stories

Explore

DIW Data Narrator

HomeExplore DataStories AnalyserStories

Back

Macroeconomics

Economic trends, policy analysis, and market dynamics

124161

Total StoriesDomainsMinutes

Search stories...Filter

The Inflation-Growth Trap

Advanced

How Germany's Economy is Stalling in a New Era of Persistent Price Pressures

Germany is caught in a perilous economic condition where inflation remains stubbornly elevated while growth stagnates. This is not simply a rebound from global...

GermanyInflationEconomic Policy+1

12min1/15/20246 sectionsRead

Decoupling and the End of the Global Playbook

Intermediate

How Germany's Export-Led Growth Model Is Being Rewritten by...

Germany's economic miracle since the 1990s was powered by globalization: frictionless trade, integrated supply chains, and surging demand from emerging...

TradeChinaExports+1

15min1/10/20247 sectionsRead

Built-in Fragilities: Germany's Silent Slowdown

Advanced

Why the Engine of Europe Is Slowing Not with a Bang, But a Persistent Whisper

Germany's economic slowdown isn't sudden or spectacular — it's silent, structural, and deeply embedded in the architecture of its economy.

ProductivityDemographicsInfrastructure

10min1/8/20246 sectionsRead

The German Growth Engine is Misfiring

Intermediate

Traditional Economic Drivers Are Faltering in a Rapidly Changing Global...

Germany's traditional economic drivers — exports and manufacturing — are faltering in a rapidly changing global environment, and there is no clear replacement...

ManufacturingExportsInnovation

12min1/5/20247 sectionsRead



Decoupling and the End of the Global Playbook

How Germany's Export-Led Growth Model Is Being Rewritten by Fragmentation, Friction, and Geopolitics

10 min read

7 sections

Intermediate

Updated 2024-01-10

Trade China Exports Geopolitics

Central Thesis

Germany's economic miracle since the 1990s was powered by globalization: frictionless trade, integrated supply chains, and surging demand from emerging markets — especially China. But in the 2020s, that model is breaking down.

Section 1: I. A Nation Built on Trade — And a Dependence on China

1 of 7 sections

Hide Data

< Previous

Next >

I. A Nation Built on Trade — And a Dependence on China

By 2019, exports accounted for 47.4% of German GDP, one of the highest shares in the OECD. China became Germany's largest single trading partner, with bilateral trade reaching over €245 billion in 2022.

In 2023, German exports to China declined by 10.2% YoY, driven by weaker demand, rising competition from Chinese EVs and solar tech, and growing political mistrust.

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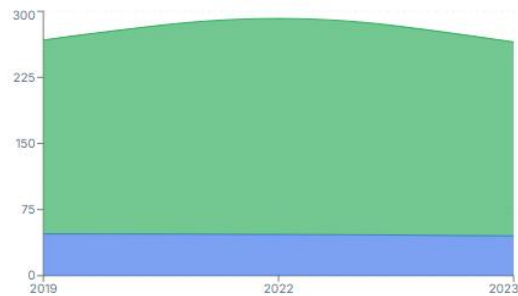


Section Data Analysis

Interactive visualizations based on the data and themes discussed in this section

Trade Dependence

Analysis of trade dependence data for this section

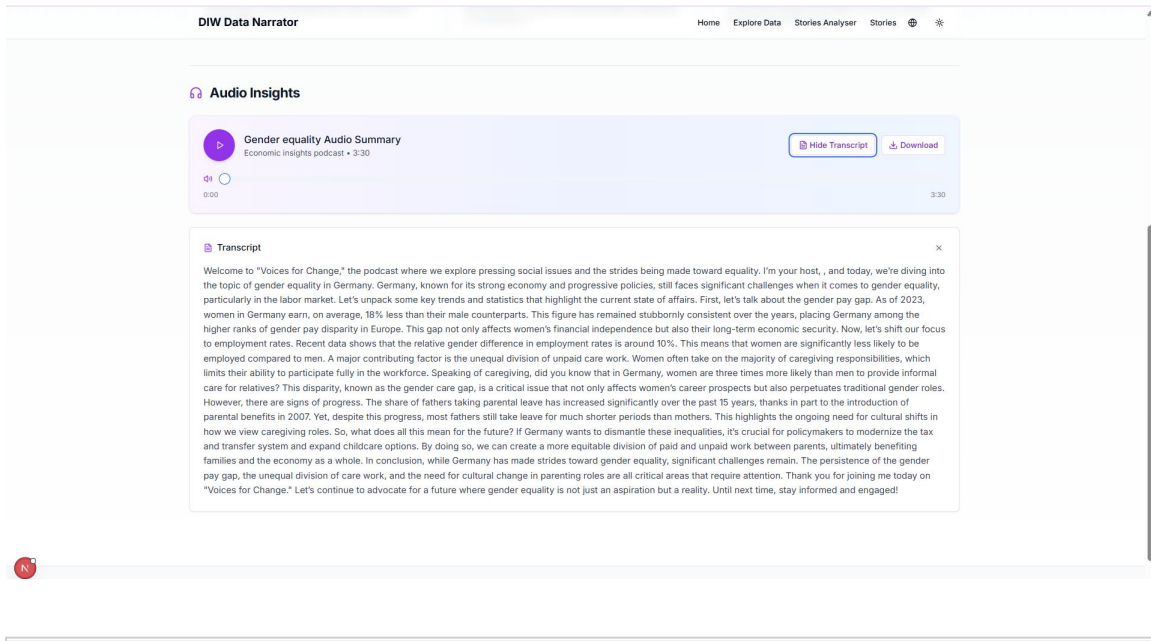


Key Insight: Exports Gdp remains relatively stable over the period.

11. Podcast Generator

- On the home page, users can click the **“Generate Podcast”** button after selecting a domain.
- The app uses a Text-to-Speech engine to generate a brief audio summary of key indicators.
- The generated podcast (.mp3) can be downloaded or played directly in the browser.
- This is useful for passive listening and accessibility.

✓ Typical Use Case: “Generate audio summary for Labour market trends in 2024.”



Backend and Frontend Setup

Backend Setup

```
cd Backend
pip install -r requirements.txt
set FLASK_APP=manage.py
flask run
```

Then visit: <http://127.0.0.1:5000/api/load-data>

✓ Terminal output: “Enhanced Flask backend with comprehensive data visualization APIs!”

To load data into PostgreSQL, use `lib/data_loader.py`. Ensure your PostgreSQL is running and configured in pgAdmin.

Frontend Setup

```
cd Frontend
npm install
npm run dev
```

Directory Notes

- `Data/`: Includes pdf/ and csv/ files.
 - `Backend/lib/data_loader.py`: Loads CSV into PostgreSQL.
 - `Backend/routes/`: Flask route definitions.
 - `Backend/services/`: Functions to query the database.
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✓ Summary

The DIW Data Narrator provides an interactive and intelligent interface to explore economic data visually, narratively, and conversationally. The system combines AI chat, chart visualizations, storytelling templates, and audio summaries to bring German economic insights to life.

Let me know if you would like this exported as a PDF or Markdown for GitHub release or academic submission.